Probability and Statistics 1
Chapter 3: Probability calculus

OUCHENE Abdelhak – 2026–2027

1 Introduction

The first two chapters described data that had already been observed. Probability does the opposite: it models what *may* happen, before any observation, and attaches to each possible outcome a number that measures how likely it is. The shift is considerable. A frequency $f_i = n_i/n$ is a fact about a sample; a probability $\mathbb{P}(A)$ is a property of a model of the experiment. The two are related — this is the whole point of the law of large numbers, which is beyond the scope of this course — but they are not the same object, and confusing them is the source of most errors in the applications.

Probability is the language of every part of computer science that deals with uncertainty. A randomised algorithm succeeds with probability $1 - \varepsilon$; a hash function collides with a probability that depends on the size of the table; a spam filter classifies an email on the basis of the words it contains; an intrusion-detection system raises an alert whose probability of being a false positive may exceed its probability of being a true one; a distributed system tolerates failures because the probability that two independent replicas fail simultaneously is the product of their individual failure probabilities. In each case, the question is not "what did I observe?" but "what is the set of things that could happen, and how likely is each of them?".

This chapter builds the vocabulary and the calculus that answer this question. It proceeds in four stages. *Combinatorics* (Section 2) counts the outcomes of an experiment when all of them are equally likely, which is the situation of passwords, PIN codes, hashes and random draws. *Events and set algebra* (Section 3) define the trial space, the events and the operations on events, and introduce the algebraic structures — algebras and σ-algebras — on which a probability can be defined. *Probability spaces* (Section 4) state the axioms of Kolmogorov and derive the properties used in every computation. *Conditional probability, total probability, Bayes' formula and independence* (Sections 5 to 7) extend the calculus to the situation where information arrives and where hypotheses must be revised in the light of evidence, which is the daily business of classification, detection and diagnosis.

Example.

1. *Randomised algorithm.* A Monte Carlo algorithm returns the correct answer with probability 0.9 and runs in one second. Repeating it three times and taking the majority answer raises the probability of correctness to 0.972, by an argument of independence developed in Section 7. This is a direct application of the calculus of this chapter.

2. *Spam filter.* An email containing the word "free" is classified as spam. The filter is correct on 80 % of the spam that contains the word and wrong on 10 % of the legitimate emails that contain it. What is the probability that an email containing "free" is really spam? It is not 80 %, nor 90 %, but a number that depends on the proportion of spam in the inbox, computed by Bayes' formula in Section 6.

3. *Intrusion detection.* An intrusion-detection system detects 99 % of the intrusions and raises a false alert on 5 % of the normal connections. Intrusions are 1 % of the traffic. Given an alert, what is

the probability that there is an intrusion? The answer, computed in Section 6, is approximately 0.167: five alerts out of six are false, which is the base-rate fallacy that every security analyst must understand.

2 Combinatorics

Combinatorics counts. When a random experiment has a finite number of equally likely outcomes, the probability of an event reduces to a ratio of cardinalities, and the whole difficulty of the problem is to count the favourable outcomes and the total outcomes correctly. The four counting rules below cover the situations that occur in practice.

2.1 The fundamental principle of counting

Definition. Fundamental principle of counting

If a first operation can be performed in n_1 ways, a second in n_2 ways, …, a k-th in n_k ways, and if the operations are performed in succession, then the number of possible sequences of k operations is the product

$$n_1 \, n_2 \cdots n_k.$$

Example.

1. *Password.* A password of length 8 uses only lowercase letters, of which there are 26. Each position is chosen independently of the others, so the number of possible passwords is

   $$26 \times 26 \times \cdots \times 26 = 26^8 = 208\,827\,064\,576 \approx 2.09 \times 10^{11}.$$

   This is the count on which password policies rest: an 8-letter lowercase password has about 2.1×10^{11} possibilities, a 12-letter one has $26^{12} \approx 9.5 \times 10^{16}$, a factor 4.5×10^5 more.

2. *IPv4 address.* An IPv4 address is a sequence of 4 bytes, each taking 256 values, hence

   $$256 \times 256 \times 256 \times 256 = 256^4 = 2^{32} \approx 4.29 \times 10^9.$$

   The exhaustion of the IPv4 space announced since the 1990s is this number being approached.

2.2 Arrangements

An arrangement is an ordered selection of p elements among n. The number of possibilities depends on whether the same element may be selected twice.

Definition. Arrangement with repetition

An **arrangement with repetition** of p elements among n is an ordered sequence of p elements chosen among n, where repetitions are allowed. Its number is

$$n^p.$$

Example.

1. *PIN code.* A PIN code of 4 digits, each between 0 and 9, allows repetitions. The number of possible PIN codes is
$$10^4 = 10\,000.$$
A trial-and-error attack at 1 attempt per second exhausts the space in less than three hours, which is why bank cards lock after three failures.

2. *Session token.* A session token of 32 hexadecimal characters, each from 16 symbols, has
$$16^{32} = 2^{128} \approx 3.4 \times 10^{38}$$
possibilities, which is the order of magnitude of the AES-128 key space and the reason why such tokens cannot be guessed.

Definition. Arrangement without repetition

An **arrangement without repetition** of p elements among n, with $p \leq n$, is an ordered sequence of p *distinct* elements chosen among n. Its number is
$$A_n^p = \frac{n!}{(n-p)!} = n(n-1)(n-2)\cdots(n-p+1).$$

Example.

1. *GPU allocation.* A cluster has 10 distinct GPUs and we must assign 3 of them to 3 distinct tasks, each GPU being used at most once. The number of allocations is
$$A_{10}^3 = 10 \times 9 \times 8 = 720.$$

2. *Committee with distinct roles.* From 20 students, we choose a president, a vice-president and a secretary, each student holding at most one position. The number of ways is
$$A_{20}^3 = 20 \times 19 \times 18 = 6840.$$

2.3 Permutations

A permutation is an arrangement of all n elements, that is, an ordering.

Definition. Permutation without repetition

A **permutation without repetition** of n objects is an ordering of these n objects. Its number is
$$A_n^n = n!.$$

Example.

1. *Sorting.* A sorting algorithm receives a list of n distinct elements. The input is one of the $n!$ possible orderings. For $n = 10$, $10! = 3\,628\,800$; for $n = 20$, $20! \approx 2.43 \times 10^{18}$. This is the number of leaves of a comparison-based sorting decision tree, hence the lower bound $\lceil \log_2(n!) \rceil \sim n \log_2 n$ on the number of comparisons.

2. *Scheduling.* A build system must order n independent compilation jobs; the number of possible orders is $n!$, reduced to a smaller number when the jobs have dependencies (a topological-sort count).

Definition. Permutation with repetition

A **permutation with repetition** of n objects, where each object may appear any number of times, is an arrangement with repetition of n elements among n. Its number is

$$n^n.$$

Example.

A function from a set of n elements to itself is exactly a sequence of n values chosen among n, hence one of the n^n permutations with repetition. For $n = 5$, there are $5^5 = 3125$ such functions, among which $5! = 120$ are bijections (permutations without repetition).

2.4 Combinations

A combination is a subset, that is, an unordered selection. Order does not matter, and repetitions are not allowed.

Definition. Combination

A **combination** of p elements among n, with $p \leq n$, is a subset of p elements chosen among n, without regard to order. Its number is

$$C_n^p = \binom{n}{p} = \frac{A_n^p}{p!} = \frac{n!}{p!\,(n-p)!}.$$

Example.

1. *Feature selection.* From 20 candidate features, we choose a subset of 5 to train a model. The order of selection is irrelevant and repetitions are impossible, so the number of subsets is

$$C_{20}^5 = \frac{20!}{5!\,15!} = 15\,504.$$

2. *Subsets of a set.* A set of n elements has C_n^p subsets of size p, hence 2^n subsets in total. For $n = 8$, there are $2^8 = 256$ subsets, which is the number of possible values of a byte when each bit indicates the presence or absence of an element.

3. *Poker hands.* A hand of 5 cards from a 52-card deck is a combination, of which there are

$$C_{52}^5 = \frac{52!}{5!\,47!} = 2\,598\,960.$$

The following table summarises the four counting rules; the two questions, *does order matter?* and *are repetitions allowed?*, determine the formula.

Rule	Formula	Order matters?	Repetition?
Arrangement with repetition	n^p	yes	yes
Arrangement without repetition	$A_n^p = n!/(n-p)!$	yes	no
Permutation	$A_n^n = n!$	yes	no
Permutation with repetition	n^n	yes	yes
Combination	$C_n^p = n!/(p!\,(n-p)!)$	no	no

Table 2.1: The four counting rules and the two questions that select among them.

Remark.

Combinations and arrangements differ only by the factor $p!$, which is the number of orderings of the selected elements: $A_n^p = p!\,C_n^p$. This factor is the only thing to remember, and it is also the most frequent source of error: choosing a committee (unordered) is not the same as choosing a president and a secretary (ordered), even when the same people are involved.

3 Random experiments and events

Combinatorics answers the question *how many?*. Probability answers the question *how likely?*. To do so, it needs a precise description of what may happen, and this description is the trial space.

Definition. Random experiment

A **random experiment** is an experiment that can be repeated under the same conditions and whose outcome cannot be predicted with certainty.

Example.

1. *Random number generator.* A pseudo-random generator is seeded with the current time and returns an integer between 1 and 6; the result cannot be predicted without knowing the seed, and the experiment can be repeated with a new seed.
2. *Network packet.* A packet arriving at a router is a random experiment: its size, its source address, its protocol cannot be predicted in advance, and the observation can be repeated packet after packet.
3. *User click.* A user is shown an advertisement and either clicks or does not; the outcome is unknown before the display, and the experiment can be repeated on another user.

Definition. Sample space (trial space)

The **sample space**, also called **trial space**, of a random experiment is the set of all its possible outcomes. It is denoted by Ω.

Example.

1. *Dice.* $\Omega = \{1, 2, 3, 4, 5, 6\}$, a finite set of 6 elements.

2. *Byte.* $\Omega = \{0,1\}^8$, the set of 256 binary strings of length 8.
3. *Latency.* $\Omega = [0, +\infty[$, the set of possible latencies in milliseconds; the sample space is infinite and uncountable.
4. *Repeated tosses.* If the experiment consists in tossing a coin until the first head, then $\Omega = \{H, TH, TTH, TTTH, \dots\}$, a countably infinite set.

Definition. Elementary outcome

An **elementary outcome** is an element ω of the sample space Ω.

Example.

1. For the dice, the elementary outcome $\omega = 3$ is the result "three".
2. For the byte, the elementary outcome $\omega = 10110101$ is one specific binary string.
3. For the latency, the elementary outcome $\omega = 42.37$ is one specific measurement.

Definition. Event

An **event** is a subset A of the sample space Ω. The set of all events, the power set of Ω, is denoted $\mathcal{P}(\Omega)$.

Example.

1. For the dice, the event $A = \{2, 4, 6\}$ is "the result is even".
2. For the latency, the event $A = [100, +\infty[$ is "the request took at least 100 ms".

Definition. Realisation of an event

An event A is **realised** if the elementary outcome ω produced by the experiment belongs to A.

Example.

1. If the dice shows 4, the event "even" is realised and the event "greater than 4" is not.
2. If a packet of 1500 bytes arrives, the event "jumbo frame" (size greater than 9000 bytes) is not realised.

Definition. Certain and impossible events

The sample space Ω itself is the **certain event**, and the empty set $\emptyset$ is the **impossible event**.

Example.

For the dice, $\Omega = \{1, \dots, 6\}$ is realised at every throw, and $\emptyset$ is realised at no throw. The event "the result is between 1 and 6 included" is certain; the event "the result is 7" is impossible.

3.1 Set algebra of events

Events are sets, and the operations of set theory translate into logical operations on events. The following definitions recall these operations in the language of probability.

Definition. Complement

The **complement** of an event A, denoted $\overline{A}$ (or A^c), is the event "A is not realised":

$$\overline{A} = \{\omega \in \Omega : \omega \notin A\}.$$

Example.

For the dice, the complement of $A = \{2, 4, 6\}$ is $\overline{A} = \{1, 3, 5\}$, the event "the result is odd".

Definition. Inclusion

The event A **implies** the event B, denoted $A \subseteq B$, if every outcome realising A also realises B.

Example.

For the dice, $\{2\} \subseteq \{2, 4, 6\}$: showing 2 implies showing an even number. The event "greater than 4" is not included in "even", since 5 is greater than 4 and odd.

Definition. Intersection

The **intersection** of two events A and B, denoted $A \cap B$, is the event realised if and only if both A and B are realised:

$$A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}.$$

Example.

For the dice, $\{2, 4, 6\} \cap \{3, 6\} = \{6\}$: the result is both even and a multiple of 3 only when it is 6.

Definition. Union

The **union** of two events A and B, denoted $A \cup B$, is the event realised if and only if at least one of A or B is realised:

$$A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}.$$

Example.

For the dice, $\{2, 4, 6\} \cup \{3, 6\} = \{2, 3, 4, 6\}$: the result is even or a multiple of 3 in four cases.

Definition. Difference and symmetric difference

The **difference** $A \setminus B$ is the event realised if A is realised and B is not:

$$A \setminus B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \notin B\} = A \cap \overline{B}.$$

The **symmetric difference** $A \triangle B$ is the event realised if exactly one of A and B is realised:

$$A \triangle B = (A \setminus B) \cup (B \setminus A) = (A \cap \overline{B}) \cup (\overline{A} \cap B).$$

Example.

For the dice, $\{2,4,6\} \setminus \{3,6\} = \{2,4\}$ and $\{2,4,6\} \triangle \{3,6\} = \{2,3,4\}$: the symmetric difference contains the outcomes belonging to exactly one of the two events.

The operations are visualised by Venn diagrams, of which Figure 3.1 gives the generic form.

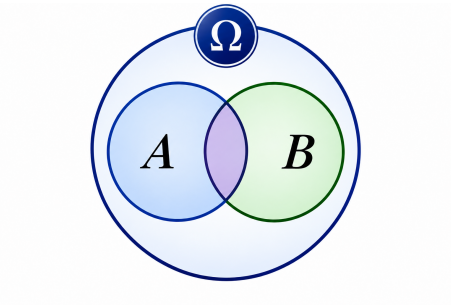


Figure 3.1: Venn diagram of two events A and B inside the sample space.

3.2 Algebra of events and σ-algebra of events

Not every family of subsets of Ω is suitable as the domain of a probability. The family must be closed under the operations that probability uses, complement and union, which leads to the following structures.

Definition. Algebra of events

A family $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is an **algebra of events** on Ω if:

(i) $\Omega \in \mathcal{A}$;

(ii) $A \in \mathcal{A} \Rightarrow \overline{A} \in \mathcal{A}$;

(iii) $A, B \in \mathcal{A} \Rightarrow A \cup B \in \mathcal{A}$.

Proposition.

If $\mathcal{A}$ is an algebra of events on Ω, then $\emptyset \in \mathcal{A}$, $\mathcal{A}$ is closed under difference and finite intersection, and it is closed under finite union of any length.

Example.

1. For $\Omega = \{1,2,3,4\}$, the family $\mathcal{A} = \{\emptyset, \{1,2\}, \{3,4\}, \Omega\}$ is an algebra: it contains Ω, is closed under complement ($\overline{\{1,2\}} = \{3,4\}$) and under union.
2. The power set $\mathcal{P}(\Omega)$ is always an algebra, and it is the largest one.
3. The trivial family $\{\emptyset, \Omega\}$ is always an algebra, and it is the smallest one.
4. For $\Omega = \{1,2,3,4\}$, the family $\{\emptyset, \{1\}, \{2,3,4\}, \Omega\}$ is an algebra generated by the partition $\{\{1\}, \{2,3,4\}\}$.

Definition. σ-algebra of events

A family $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ is a **σ-algebra** of events on Ω if it is an algebra and it is closed under countable union: for every sequence $A_1, A_2, \ldots$ of elements of $\mathcal{A}$,

$$\bigcup_i A_i \in \mathcal{A}.$$

Example.

1. On a finite space, every algebra is a σ-algebra, since the countable union of elements of a finite family is a finite union.
2. On $\Omega = \mathbb{R}$, the Borel σ-algebra, generated by the intervals, is the standard domain of probability on the real line; it contains every set that can be obtained from intervals by countable unions and complements.
3. The intersection of any family of σ-algebras on Ω is again a σ-algebra; this property is what allows one to define the σ-algebra *generated* by a family of events, as the smallest σ-algebra containing it.

Remark.

The distinction between algebra and σ-algebra is the distinction between finite experiments and infinite ones. On a finite sample space, the two notions coincide and the power set is a σ-algebra, so every subset is an event. On an infinite sample space, such as the set of latencies $[0, +\infty[$, not every subset can be assigned a probability without contradiction, and one restricts oneself to a σ-algebra, typically the Borel one. This subtlety is invisible in the exercises of this course, which are all set on finite or countable spaces, but it is the reason why the axiomatic definition of Kolmogorov is stated in terms of a σ-algebra and not of the power set.

4 Probability spaces

A probability is a function that assigns to each event a number between 0 and 1, in a way that is compatible with the set algebra of the previous section. The axioms below, due to Kolmogorov (1933), are the minimal requirements that such a function must satisfy; every property used in the applications is a consequence of them.

Definition. Probability space

A **probability space** is a triple $(\Omega, \mathcal{A}, \mathbb{P})$ where Ω is a sample space, $\mathcal{A}$ is a σ-algebra of events on Ω, and $\mathbb{P}$ is a function from $\mathcal{A}$ to $[0, 1]$ satisfying the three axioms:

(i) $\mathbb{P}(\Omega) = 1$;

(ii) $\mathbb{P}(A) \geq 0$ for every $A \in \mathcal{A}$;

(iii) for every sequence $A_1, A_2, \ldots$ of pairwise disjoint events of $\mathcal{A}$,

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i).$$

The number $\mathbb{P}(A)$ is called the **probability** of the event A.

Example.

1. *Fair dice.* $\Omega = \{1, \ldots, 6\}$, $\mathcal{A} = \mathcal{P}(\Omega)$, and $\mathbb{P}(\{i\}) = 1/6$ for every i. Axiom (i) reads $\sum_{i=1}^{6} 1/6 = 1$, axiom (ii) is obvious, and axiom (iii) follows from the additivity of the sum over disjoint subsets.
2. *Biased coin.* $\Omega = \{H, T\}$, $\mathbb{P}(\{H\}) = p$ and $\mathbb{P}(\{T\}) = 1 - p$ for some $p \in [0, 1]$. The three axioms hold for every p, which shows that a probability space is a *model* and not a property of the coin.
3. *Uniform byte.* $\Omega = \{0, 1\}^8$, $\mathbb{P}(\{\omega\}) = 1/256$ for every byte ω.

Proposition. *Elementary properties*

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. For all events $A, B \in \mathcal{A}$:

(i) $\mathbb{P}(\emptyset) = 0$;

(ii) $\mathbb{P}(\overline{A}) = 1 - \mathbb{P}(A)$;

(iii) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;

(iv) $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$.

The proofs are one-line consequences of the axioms. For (i), Ω and $\emptyset$ are disjoint and their union is Ω, so $1 = \mathbb{P}(\Omega) = \mathbb{P}(\Omega) + \mathbb{P}(\emptyset)$. For (ii), A and $\overline{A}$ are disjoint and their union is Ω, so $1 = \mathbb{P}(A) + \mathbb{P}(\overline{A})$. For (iii), decompose $A \cup B$ into the disjoint events $A \setminus B$, $A \cap B$ and $B \setminus A$, and apply additivity to each side. For (iv), write $B = A \cup (B \setminus A)$ with A and $B \setminus A$ disjoint.

Example.

1. *dice.* $\mathbb{P}(\{2, 4, 6\}) = 3/6 = 1/2$; its complement $\{1, 3, 5\}$ has probability $1 - 1/2 = 1/2$. The union $\{2, 4, 6\} \cup \{3, 6\} = \{2, 3, 4, 6\}$ has probability $4/6$, and the intersection $\{6\}$ has probability $1/6$, in agreement with the formula $3/6 + 2/6 - 1/6 = 4/6$.
2. *Byte.* The event "at least one 1" has probability $1 - \mathbb{P}(\{00000000\}) = 1 - 1/256 = 255/256$, by the complement rule.

Remark.

Property (iii) is often written in the form

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$$

only when A and B are disjoint; forgetting the correction term $-\mathbb{P}(A \cap B)$ is the most common computational error. For three events, the correction becomes

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C),$$

which is the inclusion–exclusion formula.

4.1 Construction of a probability on a finite space

On a finite sample space, a probability is entirely determined by the probabilities of the elementary outcomes, and when these outcomes are equally likely, the probability of every event reduces to a ratio of cardinalities.

Definition. Equiprobability

A probability space $(\Omega, \mathcal{P}(\Omega), \mathbb{P})$ is **equiprobable** if all the elementary outcomes have the same probability:

$$\mathbb{P}(\{\omega\}) = \frac{1}{|\Omega|} \qquad \text{for every } \omega \in \Omega.$$

In that case, for every event A,

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}.$$

Example.

1. *Fair dice.* $|\Omega| = 6$ and $\mathbb{P}(\{i\}) = 1/6$. The event "even" has $|A| = 3$ favourable outcomes, so $\mathbb{P}(A) = 3/6 = 1/2$.
2. *Uniform byte.* $|\Omega| = 256$; the event "the byte has even Hamming weight" contains 128 bytes, so its probability is $128/256 = 1/2$.
3. *Password guess.* An attacker tries one random password among the 26^8 possible 8-letter lowercase passwords; the probability of hitting the correct one is $1/26^8 \approx 4.8 \times 10^{-12}$.

Remark.

Equiprobability is a *modeling hypothesis*, not a consequence of the axioms. A dice may be fair or loaded; a random-number generator may be uniform or biased. The hypothesis must be justified by the design of the experiment, and it is the first thing to check when a probability computation gives a counter-intuitive result: many apparent paradoxes come from applying the ratio $|A|/|\Omega|$ to outcomes that are not equiprobable.

5 Conditional probability

The probability $\mathbb{P}(A)$ measures how likely A is before any information is available. In practice, information arrives: a test has failed, an alert has been raised, a packet has a given size. The probability must be updated, and the notion that does this is the conditional probability.

Definition. Conditional probability

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space and $B \in \mathcal{A}$ an event with $\mathbb{P}(B) > 0$. The **conditional probability of A given B** is the number

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Example.

1. *dice.* Given that the result is even ($B = \{2, 4, 6\}$, $\mathbb{P}(B) = 1/2$), what is the probability that it is 6? The event $A = \{6\}$ has $A \cap B = \{6\}$, so

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(\{6\})}{\mathbb{P}(\{2, 4, 6\})} = \frac{1/6}{1/2} = \frac{1}{3}.$$

   Conditioning on B has restricted the sample space to $\{2, 4, 6\}$, on which 6 is one outcome out of three.

2. *Byte.* Given that a byte has even Hamming weight, what is the probability that it is the zero byte? The event $A = \{00000000\}$ is included in the event B "even weight", so $A \cap B = A$ and

$$\mathbb{P}(A \mid B) = \frac{1/256}{1/2} = \frac{1}{128}.$$

Proposition. *The function $\mathbb{P}(\cdot \mid B)$ is a probability*

For a fixed event B with $\mathbb{P}(B) > 0$, the function $A \mapsto \mathbb{P}(A \mid B)$ satisfies the three axioms of a probability on $(\Omega, \mathcal{A})$:

(i) $\mathbb{P}(\Omega \mid B) = 1$;

(ii) $\mathbb{P}(A \mid B) \geq 0$;

(iii) for pairwise disjoint $A_1, A_2, \ldots$, $\mathbb{P}(\bigcup_i A_i \mid B) = \sum_i \mathbb{P}(A_i \mid B)$.

In particular, $\mathbb{P}(\emptyset \mid B) = 0$, $\mathbb{P}(\overline{A} \mid B) = 1 - \mathbb{P}(A \mid B)$, and $\mathbb{P}(A \cup C \mid B) = \mathbb{P}(A \mid B) + \mathbb{P}(C \mid B) - \mathbb{P}(A \cap C \mid B)$.

The verification is immediate from the definition: each property of $\mathbb{P}(\cdot \mid B)$ is the corresponding property of $\mathbb{P}$ applied to $A \cap B$, divided by the constant $\mathbb{P}(B)$.

Example.

For the dice conditioned on "even", $\mathbb{P}(\{6\} \mid B) = 1/3$ and $\mathbb{P}(\{1\} \mid B) = 0$, since 1 is not even. The complement of $\{6\}$ in the conditioned space is $\{2, 4\}$, of probability $2/3 = 1 - 1/3$, as stated.

Definition. Multiplication rule

For any two events A, B with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\,\mathbb{P}(B) = \mathbb{P}(B \mid A)\,\mathbb{P}(A).$$

More generally, for events $A_1, \dots, A_n$ with $\mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) > 0$,

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1)\,\mathbb{P}(A_3 \mid A_1 \cap A_2) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

Example.

1. *Two packets without replacement.* A batch contains 10 packets, of which 3 are corrupted. We draw two packets at random without replacement. The probability that both are corrupted is

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1) = \frac{3}{10} \times \frac{2}{9} = \frac{1}{15}.$$

   The conditional probability $2/9$ reflects the fact that, after the first corrupted packet has been drawn, only 2 corrupted packets remain among 9.

2. *Three components.* A system has three components in series; their lifetimes are modelled so that the probability that the first fails before time t is 0.1, the probability that the second fails before t given that the first has failed is 0.2, and the probability that the third fails given that the first two have failed is 0.3. The probability that all three fail before t is

$$0.1 \times 0.2 \times 0.3 = 0.006.$$

Remark.

The multiplication rule is the workhorse of the computation of composed probabilities: it turns the probability of an intersection into a product of conditional probabilities, each of which describes a simpler experiment. The order of the conditioning matters — $\mathbb{P}(A_2 \mid A_1) \neq \mathbb{P}(A_1 \mid A_2)$ in general — and the rule is symmetric only when both orders are computed: $\mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$ is the identity that leads to Bayes' formula in the next section.

6 Total probability and Bayes' formula

Conditional probability answers the question *given B, what is the probability of A?*. The two formulas of this section answer the reverse questions: *what is the probability of B, knowing its probability conditional on each of several scenarios?* and *given B, what is the probability that a particular scenario produced it?*.

6.1 Complete system of events

Definition. Complete system of events (partition)

A family $\{A_1, \dots, A_n\}$ of events of a probability space is a **complete system of events**, or a **partition** of Ω, if

$$A_i \cap A_j = \emptyset \text{ for } i \neq j, \qquad \bigcup_{i=1}^{n} A_i = \Omega.$$

Example.

1. For the dice, $\{A_1, A_2, A_3\}$ with $A_1 = \{1, 2\}$, $A_2 = \{3, 4\}$, $A_3 = \{5, 6\}$ is a complete system.
2. For the byte, the events "Hamming weight less than 4", "weight equal to 4" and "weight greater than 4" form a complete system.
3. For the email filter, $\{\text{spam}, \text{legitimate}\}$ is the complete system on which Bayes' formula is applied below.

6.2 The total probability formula

Definition. Total probability formula

Let $\{A_1, \dots, A_n\}$ be a complete system of events with $\mathbb{P}(A_i) > 0$ for every i, and let B be any event. Then

$$\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(B \mid A_i)\, \mathbb{P}(A_i).$$

The proof is a direct application of the multiplication rule and of the additivity of $\mathbb{P}$: the events $B \cap A_1, \dots, B \cap A_n$ are pairwise disjoint and their union is B, so $\mathbb{P}(B) = \sum_i \mathbb{P}(B \cap A_i) = \sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$.

Example. *Spam filter*

An inbox contains 30 % spam and 70 % legitimate email. The word "free" appears in 80 % of the spam and in 10 % of the legitimate email. The probability that a randomly chosen email contains the word "free" is

$$\mathbb{P}(\text{free}) = \mathbb{P}(\text{free} \mid \text{spam})\,\mathbb{P}(\text{spam}) + \mathbb{P}(\text{free} \mid \text{legit})\,\mathbb{P}(\text{legit}) = 0.8 \times 0.3 + 0.1 \times 0.7 = 0.24 + 0.07 = 0.31.$$

The formula decomposes a probability that is difficult to observe directly into two probabilities that are easy to estimate on labelled data.

Example. *Intrusion detection*

Intrusions are 1 % of the connections. The detector raises an alert on 99 % of the intrusions and on 5 % of the normal connections. The probability of an alert is

$$\mathbb{P}(\text{alert}) = 0.99 \times 0.01 + 0.05 \times 0.99 = 0.0099 + 0.0495 = 0.0594.$$

Only about 6 % of the connections raise an alert, even though the detector is correct on 99 % of the intrusions, because the normal connections are 99 times more numerous. This is the first half of the base-rate effect; the second half is Bayes' formula.

6.3 Bayes' formula

Definition. Bayes' formula

Let $\{A_1, \dots, A_n\}$ be a complete system of events with $\mathbb{P}(A_i) > 0$ for every i, and let B be an

event with $\mathbb{P}(B) > 0$. Then for every i,

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i)\,\mathbb{P}(A_i)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B \mid A_i)\,\mathbb{P}(A_i)}{\displaystyle\sum_{j=1}^{n} \mathbb{P}(B \mid A_j)\,\mathbb{P}(A_j)}.$$

Bayes' formula is the multiplication rule written in both orders: $\mathbb{P}(A_i \cap B) = \mathbb{P}(B \mid A_i)\mathbb{P}(A_i) = \mathbb{P}(A_i \mid B)\mathbb{P}(B)$, and the denominator is the total probability of B.

Example. *Spam filter, continued*

With the data of the previous example, the probability that an email containing "free" is spam is

$$\mathbb{P}(\text{spam} \mid \text{free}) = \frac{0.8 \times 0.3}{0.31} = \frac{0.24}{0.31} \approx 0.774.$$

About 77 % of the emails containing "free" are spam. The prior probability of spam was 30 %; the observation of the word has raised it to 77 %, but not to 80 %, because legitimate emails also contain the word.

Example. *Intrusion detection, continued*

With the data of the previous example, the probability that an alert corresponds to a real intrusion is

$$\mathbb{P}(\text{intrusion} \mid \text{alert}) = \frac{0.99 \times 0.01}{0.0594} = \frac{0.0099}{0.0594} \approx 0.167.$$

Five alerts out of six are false, even though the detector is correct on 99 % of the intrusions. The reason is the base rate: intrusions are 1 % of the traffic, so the 5 % false-positive rate applied to the 99 % of normal traffic produces far more alerts than the 99 % detection rate applied to the 1 % of intrusions.

The two examples are summarised by the contingency table below, which makes the arithmetic transparent and is the recommended way of solving any Bayes exercise with two hypotheses.

	Alert	No alert	Total
Intrusion	99	1	100
No intrusion	495	9405	9900
Total	594	9406	10000

Table 6.1: Contingency table of the intrusion-detection example on 10 000 connections. The conditional probability $\mathbb{P}(\text{intrusion} \mid \text{alert}) = 99/594 \approx 0.167$ is read directly in the first column.

Remark.

Bayes' formula is not a different rule from the total probability formula: it is the same computation read in the other direction. The difficulty of the exercises is never in the formula, but in identifying the hypotheses A_i, the observation B, and which of the three probabilities $\mathbb{P}(A_i)$, $\mathbb{P}(B \mid A_i)$, $\mathbb{P}(A_i \mid B)$ is known and which is asked. Writing the contingency table, as above, resolves this difficulty in every two-hypothesis exercise.

Remark. *Base-rate fallacy*

The comparison between $\mathbb{P}(\text{intrusion} \mid \text{alert}) \approx 0.167$ and the sensitivity $\mathbb{P}(\text{alert} \mid \text{intrusion}) = 0.99$ is the base-rate fallacy: confusing the probability of the evidence given the hypothesis with the probability of the hypothesis given the evidence. The two numbers differ by two orders of magnitude in this example, and the error is systematic in medical diagnosis, security alerts and fraud detection, where the event of interest is rare. The prior probability is not a detail; it is the main term of Bayes' formula when the evidence is rare.

7 Independence

Two events are independent when the realisation of one does not change the probability of the other. This is the formal counterpart of the intuition that two experiments that do not influence each other should be modelled by probabilities that multiply.

Definition. Independent events

Two events A and B of a probability space are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

Example.

1. *Two dice.* Two fair dice are thrown. The event "the first dice shows 6" and the event "the second dice shows 6" are independent: $\mathbb{P}(A \cap B) = 1/36 = \mathbb{P}(A)\mathbb{P}(B)$. The event "the sum is 7" is not independent of the first dice, since knowing the first value determines the second when the sum is 7.

2. *Two replicas.* A distributed system stores two replicas of a file on two independent disks; each disk fails during a year with probability 0.02. The probability that both fail during the same year is $0.02^2 = 0.0004$, which is the reason why replication is used.

3. *Hash collision.* A hash function sends n keys to m slots uniformly and independently. The probability that two given keys collide is $1/m$; the probability that no two of the n keys collide is the birthday problem, computed by the multiplication rule over independent pairs.

Proposition. *Independence and complementation*

If A and B are independent, then so are the pairs $(A, \overline{B})$, $(\overline{A}, B)$ and $(\overline{A}, \overline{B})$.

The proof for $(A, \overline{B})$ uses the additivity of $\mathbb{P}$ on the partition $\{B, \overline{B}\}$:

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap \overline{B}),$$

hence $\mathbb{P}(A \cap \overline{B}) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)(1 - \mathbb{P}(B)) = \mathbb{P}(A)\mathbb{P}(\overline{B})$. The other two cases follow by symmetry.

Example.

If the two replicas of the previous example are independent in their failures, the probability that the first fails and the second survives is $0.02 \times 0.98 = 0.0196$, and the probability that at least one

survives is $1 - 0.0004 = 0.9996$. The complement of "both fail" is "at least one survives", and it is computed by the complement rule, not by adding the individual survival probabilities.

Remark. *Independence is not incompatible events*

Two events are *incompatible* when they cannot occur simultaneously, that is, $A \cap B = \emptyset$; they are *independent* when the occurrence of one does not modify the probability of the other, that is, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. For events of positive probability, the two notions are mutually exclusive: if $A \cap B = \emptyset$, then $\mathbb{P}(A \cap B) = 0 \neq \mathbb{P}(A)\mathbb{P}(B) > 0$, so incompatible events are *never* independent. The confusion between the two is the most frequent conceptual error in the first exercises of probability.

Definition. Mutual independence

A family $\{A_1, \dots, A_n\}$ of events is **mutually independent** if, for every subset $I \subseteq \{1, \dots, n\}$ of size at least 2,

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i).$$

Example.

1. *Three components.* Three components fail independently with probabilities $0.1, 0.2, 0.3$. The probability that all three fail is $0.1 \times 0.2 \times 0.3 = 0.006$; the probability that none fails is $0.9 \times 0.8 \times 0.7 = 0.504$; the probability that at least one fails is $1 - 0.504 = 0.496$. Pairwise independence would not be enough to justify the first computation; mutual independence is the hypothesis used.

2. *Triple of events.* It is possible to construct three events that are pairwise independent but not mutually independent: for $\Omega = \{1, 2, 3, 4\}$, the events $\{1, 2\}$, $\{1, 3\}$ and $\{1, 4\}$ with the uniform probability are pairwise independent (each intersection has probability $1/4 = 1/2 \times 1/2$) but their triple intersection $\{1\}$ has probability $1/4 \neq 1/8$. This shows that mutual independence is strictly stronger than pairwise independence.

Definition. Independent experiments

A sequence of random experiments is **independent** if, for every choice of an event in each experiment, the events obtained are mutually independent.

Example.

1. *Repeated requests.* Requests arriving at a load balancer are modelled as independent experiments; the probability that three consecutive requests come from the same region is the product of the three marginal probabilities, by independence.

2. *Repeated algorithm runs.* A randomised algorithm is run n times independently; the probability that it fails on all n runs is $(1 - p)^n$, where p is its success probability on a single run. For $p = 0.9$ and $n = 3$, this is 0.001, which is the basis of the repetition technique mentioned in the introduction.

Proposition. *Composed probabilities*

If $A_1, \ldots, A_n$ are mutually independent, then

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2) \cdots \mathbb{P}(A_n).$$

More generally, the multiplication rule of Section 5 takes the simpler form

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2) \cdots \mathbb{P}(A_n)$$

when the A_i are mutually independent, since $\mathbb{P}(A_i \mid A_1 \cap \cdots \cap A_{i-1}) = \mathbb{P}(A_i)$.

Example.

A backup system has three independent layers: a primary server that fails with probability 0.05, a replica that fails with probability 0.05, and an off-site backup that fails with probability 0.10. The probability that all three fail is $0.05 \times 0.05 \times 0.10 = 0.00025$, and the probability that at least one layer survives is $1 - 0.00025 = 0.99975$. The computation is a direct consequence of mutual independence; it is the argument used to justify redundancy in any distributed system.

Remark.

Independence is a *hypothesis on the model*, not a property that can be verified on a single experiment. Two events may be independent by design (two dice, two independent replicas) or correlated by the physics of the system (two replicas on the same disk are not independent). Assuming independence without justification is the second most frequent error of the applications, after the base-rate fallacy: it leads to the same underestimation of risk in both cases.

8 Common mistakes

1. **Confusing independent and incompatible.** Incompatible events cannot occur together; independent events do not influence each other. For events of positive probability, they are mutually exclusive notions.

2. **Forgetting the correction term in the union.** $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, not $\mathbb{P}(A) + \mathbb{P}(B)$, unless A and B are disjoint. The error doubles the probability of the union when the events overlap.

3. **Confusing $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$.** The probability of a positive test given the disease is not the probability of the disease given a positive test. The two are related by Bayes' formula, and they can differ by orders of magnitude.

4. **Ignoring the prior in Bayes' formula.** The denominator $\mathbb{P}(B)$ is not a normalisation constant to be forgotten: it is the total probability of the evidence, and it is what incorporates the base rate. Dropping it produces the base-rate fallacy.

5. **Assuming independence without justification.** Two failures on the same machine, two requests from the same user, two measurements from the same sensor are not independent. The multiplication $\mathbb{P}(A)\mathbb{P}(B)$ is valid only when independence is part of the model.

6. **Using the wrong sample space.** The probability of an event is defined relative to a sample space. Changing the sample space (for example, conditioning on an event) changes the probabilities, and a computation that mixes two sample spaces is meaningless.

7. **Confusing arrangement and combination.** Choosing a president and a secretary is an arrangement (A_n^2); choosing a committee is a combination (C_n^2). The factor 2! that separates the two is the source of many wrong answers in counting exercises.

8. **Applying the ratio $|A|/|\Omega|$ to non-equiprobable outcomes.** The formula $\mathbb{P}(A) = |A|/|\Omega|$ is valid only when the elementary outcomes are equiprobable. A loaded dice, a biased generator or a non-uniform hash function invalidates it.

9 Summary

- *Combinatorics.* The fundamental principle gives $n_1 n_2 \cdots n_k$ sequences. Arrangements with repetition: n^p. Arrangements without repetition: $n!/(n-p)!$. Permutations: $n!$. Permutations with repetition: n^n. Combinations: $n!/(p!(n-p)!)$. The two questions that select the formula are: does order matter? are repetitions allowed?

- *Events and set algebra.* A random experiment has a sample space Ω; an event is a subset of Ω; the operations on events are complement, inclusion, intersection, union, difference and symmetric difference. An algebra of events is closed under complement and finite union; a σ-algebra is also closed under countable union.

- *Probability space.* A triple $(\Omega, \mathcal{A}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$, $\mathbb{P}(A) \geq 0$ and σ-additivity. Consequences: $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}(\overline{A}) = 1 - \mathbb{P}(A)$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, and $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$. On a finite equiprobable space, $\mathbb{P}(A) = |A|/|\Omega|$.

- *Conditional probability.* $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ for $\mathbb{P}(B) > 0$. Multiplication rule: $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A)$, extended to n events by $\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\mathbb{P}(A_2 \mid A_1) \cdots$.

- *Total probability and Bayes.* For a complete system $\{A_i\}$ and an event B: $\mathbb{P}(B) = \sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$, and $\mathbb{P}(A_i \mid B) = \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)/\mathbb{P}(B)$. The second formula is the first read backwards; the contingency table resolves every two-hypothesis exercise.

- *Independence.* A and B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$; then so are $(A, \overline{B})$, $(\overline{A}, B)$ and $(\overline{A}, \overline{B})$. A family is mutually independent if the product rule holds for every subfamily. Independent experiments produce mutually independent events, and the multiplication rule then simplifies to a product.

- The two conceptual pitfalls of the chapter are the confusion between independent and incompatible events, and the base-rate fallacy — the confusion between $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$.

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